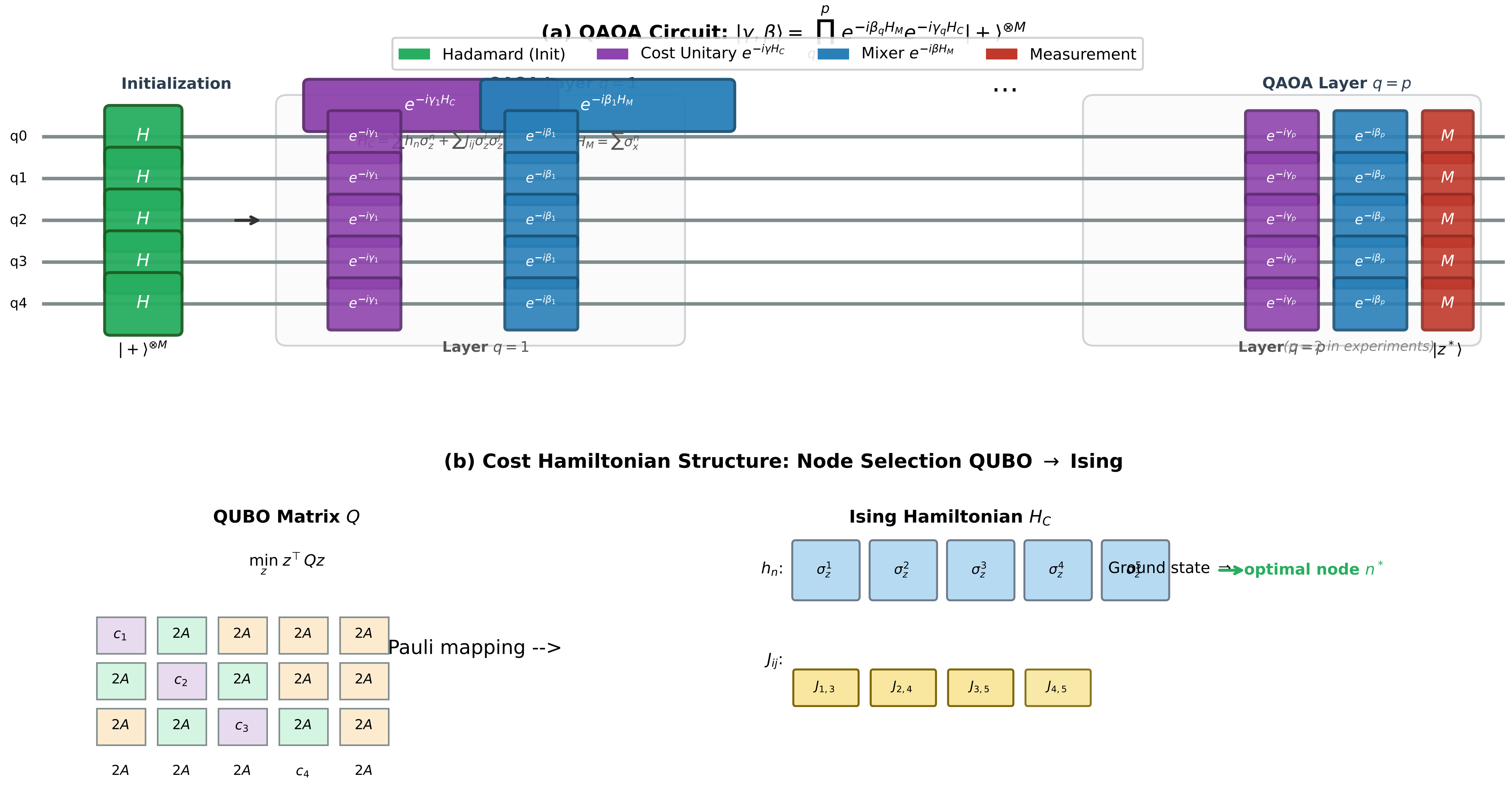

Fig QC-2: QAOA circuit for optimal node selection. The depth-p QAOA alternates between the cost unitary $e^{-i\gamma_q H_C}$ (encoding the QUBO objective) and the mixer $e^{-i\beta_q H_M}$ (enabling amplitude redistribution). Measuring the final state in the computational basis yields the optimal node bitstring z^*.